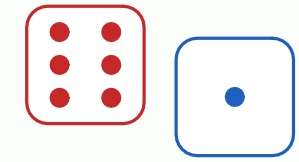
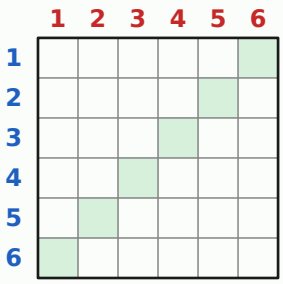
Chapter 25

Probability and Bayes

count the squares



two dice = 36 possible results, all equally likely



the 6 ways to roll a total of 7: 6 out of 36

Probability sounds like guesswork, but most of it is **counting**. List every possible result, shade the ones you care about, and the probability is **shaded ÷ total**.

This chapter uses one picture for almost everything: the **6 × 6 grid of two dice**. With it you can **see** what "not", "and", "or" and "given that" mean, and then the lecture's famous puzzle (only 15% of doctors got it right) becomes a matter of counting little people.

Goes with **Lecture 8** (sample space, complements, joint, exclusive and conditional probability, independence, the urn, Bayes' theorem, the mammogram).

Probability = shaded squares ÷ all squares

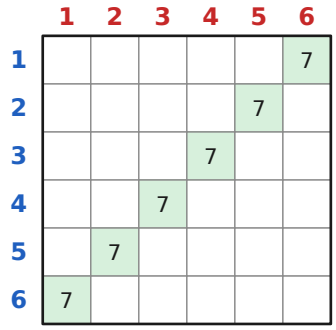
RULE: probability

For equally likely outcomes: $p(E) = \dfrac{\text{number of outcomes in } E}{\text{total number of outcomes}}$.

$0 \le p(E) \le 1$: 0 means impossible, 1 means certain. The probabilities of all the possible outcomes add up to 1.

Complement: $p(\text{not } E) = p(E') = 1 - p(E)$.

Roll a red die and a blue die. Every square below is one possible result: column = red die, row = blue die. The whole grid is the **sample space** S (every possible outcome). A set of outcomes you're interested in is an **event** E, a subset of S (Chapter 6).

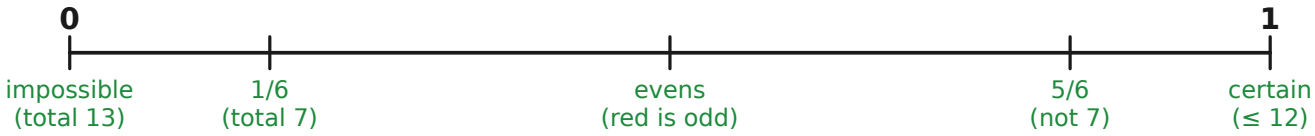


E = "the total is 7"

$$p(E) = \frac{\text{squares in } E}{\text{squares in } S} = \frac{6}{36} = \frac{1}{6}$$

read p(E) as "the probability of E"

Probabilities always sit between 0 and 1 (or 0% and 100%):



Rule 1, complements: the squares **not** in E are everything else, so

$$p(\text{not } E) = 1 - p(E)$$ ← *not a total of 7: 1 − 1/6 = 5/6*

Handy when "not" is easier to count:
$p(\text{at least one six}) = 1 - p(\text{no sixes}) = 1 - \frac{25}{36} = \frac{11}{36}$.

AND is the overlap, OR is both stripes

RULE: AND and OR

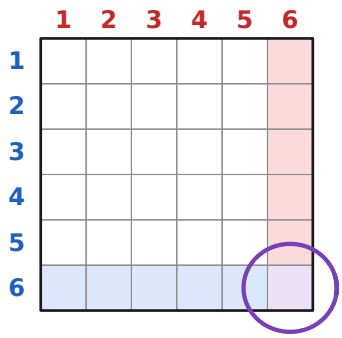
$p(A \cap B)$ = the probability that A **and** B both happen (the overlap).

$p(A \cup B)$ = the probability that A **or** B happens (either one, or both).

General addition rule, always true:
$p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Mutually exclusive events can't both happen: $p(A \cap B) = 0$, so $p(A \cup B) = p(A) + p(B)$.

Let E = "red die shows 6" (a column) and F = "blue die shows 6" (a row):



E (red column): 6 squares
F (blue row): 6 squares
both (purple corner): 1 square

$p(E \cap F) = \frac{1}{36}$ $\quad$ $p(E \cup F) = \frac{6+6-1}{36} = \frac{11}{36}$

AND (both happen, the **joint** probability) is the **overlap**: $\cap$, the intersection from Chapter 6. OR (at least one happens) is **everything shaded**: $\cup$, the union. Counting the column (6) and the row (6) counts the corner **twice**, so subtract it once:

$$p(E \cup F) = p(E) + p(F) - p(E \cap F)$$ ← *Lecture 8's Rule 3*

Mutually exclusive events can't happen together: "total is 2" and "total is 12" have no squares in common, so $p(E \cap F) = 0$ (Rule 2) and the OR rule becomes simply $p(E) + p(F) = \frac{1}{36} + \frac{1}{36}$.

"Given that": zoom in, and the stripe becomes the world

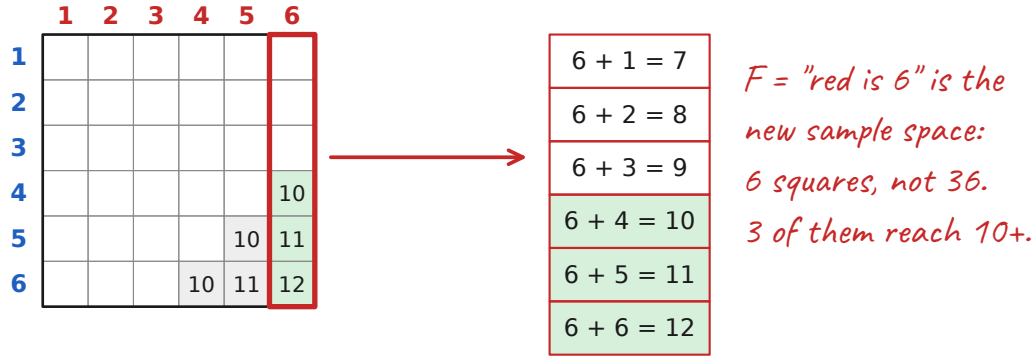
RULE: conditional probability

$p(A \mid B)$ = the probability of A **given that** B has happened. B becomes the whole sample space.

$p(A \mid B) = \dfrac{p(A \cap B)}{p(B)}$ (as long as $p(B) \neq 0$).

Multiplication rule (the same rule turned round):
$p(A \cap B) = p(B) \times p(A \mid B) = p(A) \times p(B \mid A)$.

Your friend rolls, peeks, and says "the red die is a 6". What's the chance the total is 10 or more **now**? The bar $\mid$ is read "**given**":
$p(E \mid F)$ = probability of E given that F has happened.



Without the peek, $p(10+) = \frac{6}{36} = \frac{1}{6}$. With it, $p(10+ \mid \text{red } 6) = \frac{3}{6} = \frac{1}{2}$. New information **changes** the probability. Lecture 8: **"F is the new S!"** As a formula, count the overlap inside F, as a fraction of F:

$$p(E \mid F) = \frac{p(E \cap F)}{p(F)} = \frac{3/36}{6/36} = \frac{1}{2} \leftarrow \text{Rule 4}$$

Rearranged: $p(E \cap F) = p(F) \times p(E \mid F)$: the chance of both = the chance of the first × the chance of the second **once the first has happened**.

Independent: when the peek tells you nothing

RULE: independent events

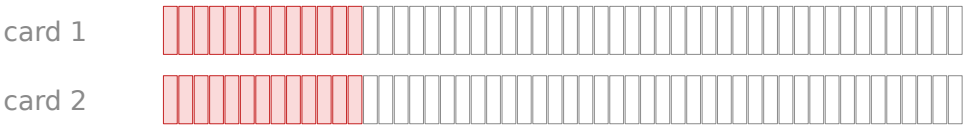
A and B are **independent** when knowing that one happened doesn't change the probability of the other: $p(A \mid B) = p(A)$.

Then, and only then: $p(A \cap B) = p(A) \times p(B)$.

Test: work out $p(A) \times p(B)$ and compare it with $p(A \cap B)$. Equal: independent. Not equal: dependent.

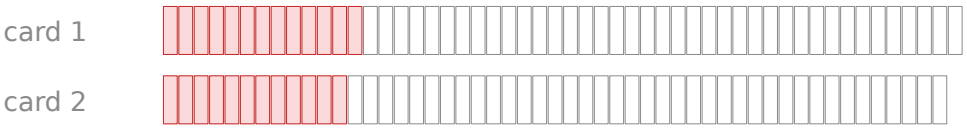
Two events are **independent** if knowing one doesn't change the other: $p(E \mid F) = p(E)$. Then the multiplication rule becomes simply $p(E \cap F) = p(E) \times p(F)$. Lecture 8's cards: **draw two cards; what's the chance both are hearts?**

put the first card back and shuffle



$= \frac{1}{16}$

keep the first card out



$= \frac{1}{17}$

With replacement the deck is identical for card 2: **independent**, $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16} = 0.0625$. Without replacement one heart has gone: **dependent**, $\frac{1}{4} \times \frac{12}{51} = \frac{3}{51} = \frac{1}{17} \approx 0.0588$.

Sun and red cars (independent): $p(\text{sun}) = 0.4$, $p(\text{red car}) = 0.2$, so $p(\text{both}) = 0.08$. Sun and umbrellas (dependent): the weather obviously changes your chance of seeing an umbrella, so you **can't** just multiply.

Exclusive or independent? The rules side by side

These two words sound alike and get mixed up constantly. They mean very different things:

	Mutually exclusive	Independent
meaning	A and B can't both happen	one happening doesn't change the chance of the other
AND	$p(A \cap B) = 0$	$p(A \cap B) = p(A)\,p(B)$
OR	$p(A \cup B) = p(A) + p(B)$	$p(A \cup B) = p(A) + p(B) - p(A)\,p(B)$
GIVEN	$p(A \mid B) = 0$	$p(A \mid B) = p(A)$
dice example	total is 2 / total is 12	red shows 6 / blue shows 6

RULE: always true, for any two events

General addition rule: $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Conditional probability: $p(A \mid B) = \dfrac{p(A \cap B)}{p(B)}$, so $p(A \cap B) = p(B)\,p(A \mid B)$.

The two special cases in the table are these general rules with $p(A \cap B)$ replaced by 0 or by $p(A)\,p(B)$.

RULE: exclusive events are never independent

If A and B are mutually exclusive and both have non-zero probability, they **cannot** be independent: $p(A \cap B) = 0$ but $p(A)\,p(B) > 0$.

Knowing A happened tells you B definitely **didn't**: the biggest change of all.

Check on the dice: "red shows 1" and "red shows 2" are exclusive. $p(\text{both}) = 0$, but $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$. Not equal, so they are dependent.

✗ Common mistakes

✗ $p(A \cup B) = p(A) + p(B)$

only drop the overlap when A and B are mutually exclusive

✓ $p(A \cup B) = p(A) + p(B) - p(A \cap B)$

✗ $p(A \cap B) = p(A)\,p(B)$ ✓ $p(A \cap B) = p(B)\,p(A \mid B)$

multiply the plain probabilities only when A and B are independent

✗ $p(A \mid B) = p(B \mid A)$ ✓ $p(A \mid B) = \dfrac{p(B \mid A)\,p(A)}{p(B)}$

a conditional can't simply be turned round (Bayes, later in the chapter)

The urn: a table does the counting

RULE: reading a two-way table

Joint $p(A \cap B)$: one cell ÷ the grand total.

Marginal $p(A)$: a row or column total ÷ the grand total.

Conditional $p(A \mid B)$: the cell ÷ the total of the row or column for B (zoom into B).

Any missing cell = its row (or column) total minus the other cells in that row (or column).

Lecture 8's urn holds 100 balls, each Red, Green or Blue and Large or Small. A **two-way table** counts every combination; the totals in the **margins** give the **marginal** probabilities:

	Red	Green	Blue	total
Large	5	30	15	**50**
Small	15	20	15	**50**
	20	**50**	**30**	**100**

Joint: $p(R \cap L) = \frac{5}{100} = 0.05$ (one cell). Marginal: $p(R) = \frac{20}{100} = 0.20$ (a margin total). Conditional: told the ball is red, **zoom into the red column** (20 balls) and it's the new world:

$$p(L \mid R) = \frac{5}{20} = 0.25 \qquad p(S \mid R) = \frac{15}{20} = 0.75$$

$$p(L \mid G) = \frac{30}{50} = 0.60$$ ← a different answer for green

Before the peek, large was 50-50. Knowing the colour changes the odds for size, so colour and size are **dependent**.

Find the missing probability

PROCEDURE: partial-information problems

1. Write down every probability you are given, in symbols: $p(A) = 0.5$, $p(A \cup B) = 0.7$, ...
2. Pick the rule that links them: the addition rule (OR), the conditional / multiplication rule (GIVEN), or $p(A \cap B) = p(A)\,p(B)$ if you're told they're independent.
3. Substitute and solve for the unknown. A Venn diagram or a two-way table helps you see it.
4. Sense-check: every probability must be between 0 and 1.

Example: $p(A) = 0.5$, $p(B) = 0.4$, $p(A \cup B) = 0.7$. Find $p(A \cap B)$. Are A and B independent?

$$0.7 = 0.5 + 0.4 - p(A \cap B) \;\Rightarrow\; p(A \cap B) = 0.2$$

addition rule

$$p(A)\,p(B) = 0.5 \times 0.4 = 0.2 = p(A \cap B)$$ ← equal, so independent ✓

Example: $p(A \mid B) = 0.3$ and $p(B) = 0.6$. Find $p(A \cap B)$.

$$p(A \cap B) = p(B)\,p(A \mid B) = 0.6 \times 0.3 = 0.18$$

multiplication rule

Example: A and B are independent, $p(A) = 0.3$ and $p(A \cup B) = 0.65$. Find $p(B)$.

$$0.65 = 0.3 + p - 0.3p$$ ← addition rule with $p(A \cap B) = 0.3p$

$$0.35 = 0.7p \;\Rightarrow\; p(B) = 0.5$$

Example: 200 people: 40% wear glasses, 15% are left-handed, and 30% of the glasses-wearers are left-handed. Complete the table. Are glasses and left-handedness independent?

Turn each percentage into a count, **of the right group**: glasses $0.4 \times 200 = 80$; left-handed $0.15 \times 200 = 30$; left-handed glasses-wearers $0.3 \times 80 = 24$ (30% **of the 80**). Every other cell is a subtraction from a total:

	left-handed	right-handed	total
glasses	**24**	80 − 24 = 56	**80**
no glasses	30 − 24 = 6	120 − 6 = 114	200 − 80 = 120
total	**30**	200 − 30 = 170	**200**

$$p(G \cap L) = \frac{24}{200} = 0.12, \qquad p(G)\,p(L) = 0.4 \times 0.15 = 0.06$$

$0.12 \neq 0.06$, so they are **dependent** (in this made-up group, glasses-wearers are more likely to be left-handed). Zoom in, as before: $p(G \mid L) = \frac{24}{30} = 0.8$, much more than $p(G) = 0.4$.

Bayes: turning a conditional round

RULE: Bayes' theorem

$p(A \mid B)$ and $p(B \mid A)$ are **different** numbers: same overlap on top, different crowd underneath.

To turn one into the other: $p(A \mid B) = \dfrac{p(B \mid A)\, p(A)}{p(B)}$.

If $p(B)$ isn't given, build it from the law of total probability (two sections on).

Lecture 8's school: 100 children. Two questions that **sound** the same but aren't:

	boy	girl	total
dark hair	30	36	66
light hair	25	9	34
total	55	45	100

light hair: zoom into the row of 34

girl: zoom into the column of 45

If you see a light-haired child, how likely is it a girl? Zoom into the light row: $p(\text{girl} \mid \text{light}) = \frac{9}{34} = 0.26$.

If you see a girl, how likely is light hair? Zoom into the girl column: $p(\text{light} \mid \text{girl}) = \frac{9}{45} = 0.20$.

Same 9 children on top, **different crowd underneath**. **Bayes' theorem** converts one into the other using the marginal probabilities, when you don't have the whole table:

$$p(a \mid b) = \frac{p(b \mid a)\, p(a)}{p(b)}$$

swap what's known and what's asked, then correct by the ratio of the crowd sizes

$$p(\ell \mid g) = \frac{p(g \mid \ell)\, p(\ell)}{p(g)} = \frac{0.26 \times 0.34}{0.45} = 0.20$$

ℓ = light, g = girl ✓

Tree diagrams and the law of total probability

RULE: tree diagrams

Split first by one event (A or A′), then split each branch by the second event (B or B′). The second-level branches carry **conditional** probabilities such as $p(B \mid A)$.

Multiply along a path to get the probability of that whole path: $p(A \cap B) = p(A)\,p(B \mid A)$.

Add between the paths that end in the outcome you want. The branches leaving any point add up to 1.

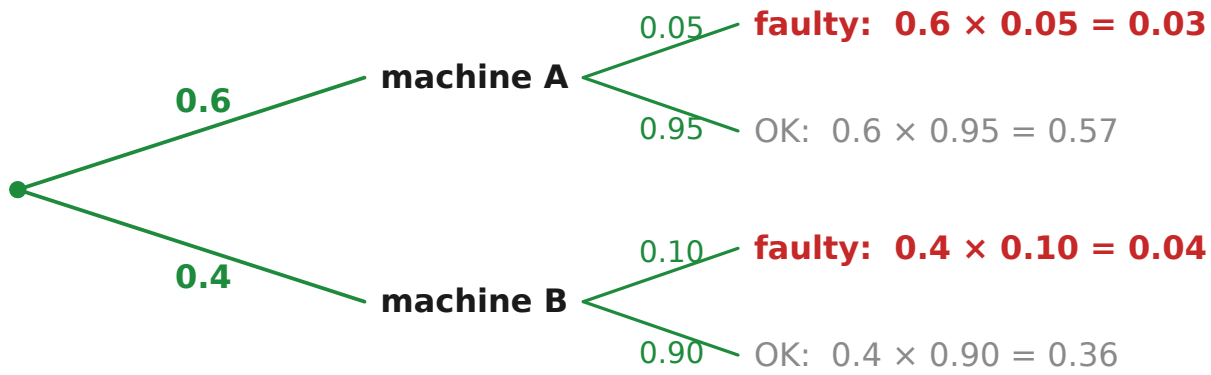
RULE: law of total probability

$p(B) = p(B \mid A)\,p(A) + p(B \mid A')\,p(A')$: every way B can happen, added up.

Bayes' theorem uses it for the bottom line:

$$p(A \mid B) = \frac{p(B \mid A)\,p(A)}{p(B \mid A)\,p(A) + p(B \mid A')\,p(A')}.$$

Example: machine A makes 60% of a factory's bolts and machine B the other 40%. 5% of A's bolts are faulty and 10% of B's. What fraction of all bolts is faulty? A bolt is faulty: what's the chance machine A made it?



$$p(F) = 0.05 \times 0.6 + 0.10 \times 0.4 = 0.03 + 0.04 = 0.07$$

add the two red paths

$$p(A \mid F) = \frac{p(F \mid A)\, p(A)}{p(F)} = \frac{0.03}{0.07} = \frac{3}{7} \approx 0.43$$

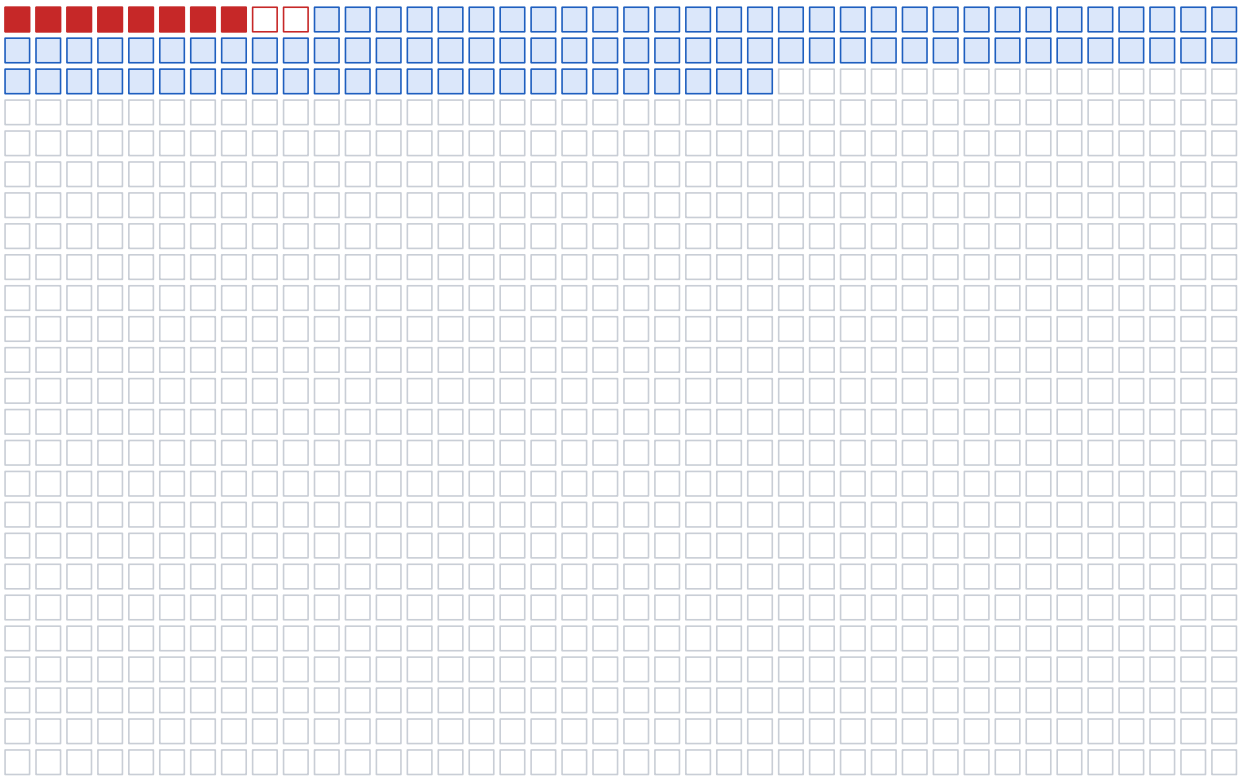
Bayes: the A path as a share of all faulty paths

Machine A makes most of the bolts, yet a faulty bolt is more likely to have come from B ($\frac{4}{7}$): B's higher fault rate wins. The mammogram puzzle next is the same calculation.

⚄ The mammogram puzzle, as 1000 people

1% of women aged 40 who are screened have breast cancer. 80% of those get a positive mammogram. 9.6% of women without cancer also get a positive. A woman tests positive. What's the chance she has cancer?

Guess first! Almost certain? Fairly likely? About half? (Lecture 8: only 15% of doctors got it right.) Now picture **1000 women**, one square each:



Step by step: 1% of 1000 = **10 have cancer**; 80% of them = **8 test positive**. The other 990 are healthy, but 9.6% of 990 = 95 test positive anyway. So **103 positives, only 8 of whom are ill**:

$$p(\text{cancer} \mid +) = \frac{8}{103} \approx 0.078 = 7.8\% \leftarrow \textit{fairly unlikely!}$$

The healthy group's chance of a false alarm is small, but **there are so many more of them** that their false alarms swamp the real cases. Bayes' theorem gives the same answer, with $p(+)$ built from both groups by the law of total probability (Lecture 8's alternative formula):

$$p(c \mid m) = \frac{p(m \mid c)\, p(c)}{p(m \mid c)p(c) + p(m \mid c')p(c')} = \frac{0.8 \times 0.01}{0.8 \times 0.01 + 0.096 \times 0.99}$$

$$= \frac{0.008}{0.10304} \approx 0.078$$

Vocabulary: $p(\text{cancer}) = 1\%$ is the **prior** (belief before the evidence); $p(\text{cancer} \mid +) = 7.8\%$ is the **posterior** (belief after). The positive test raised the risk almost eightfold, but it's still far from certain. That's the scientific method in miniature: update your beliefs as evidence arrives.

Lecture 8's extension question asks whether $p(a_1 \mid b) = \frac{p(a_1 \cap b)}{p(a_1 \cap b) + p(a_2 \cap b)}$ when a_1 and a_2 can't both happen but one of them must. **Yes:** the bottom is just every way b can happen, split into the a_1 part and the a_2 part. It's exactly the counting we did: $\frac{8}{8+95}$, with a_1 = cancer, a_2 = no cancer, b = positive.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Probability	$p(E) = \dfrac{\text{outcomes in } E}{\text{all outcomes}}$ equally likely outcomes; always between 0 and 1
Complement	$p(E') = 1 - p(E)$
Addition rule (OR)	$p(A \cup B) = p(A) + p(B) - p(A \cap B)$ always true
Mutually exclusive	$p(A \cap B) = 0, \qquad p(A \cup B) = p(A) + p(B)$ can't both happen
Conditional	$p(A \mid B) = \dfrac{p(A \cap B)}{p(B)}$ B becomes the new sample space
Multiplication rule (AND)	$p(A \cap B) = p(B)\,p(A \mid B) = p(A)\,p(B \mid A)$ always true
Independent	$p(A \cap B) = p(A)\,p(B), \qquad p(A \mid B) = p(A)$ test: compare $p(A)p(B)$ with $p(A \cap B)$
Exclusive vs independent	exclusive events with non-zero probabilities are never independent
Tree diagram	multiply along the branches, add between the paths; branches from a point add to 1
Total probability	$p(B) = p(B \mid A)\,p(A) + p(B \mid A')\,p(A')$
Bayes' theorem	$p(A \mid B) = \dfrac{p(B \mid A)\,p(A)}{p(B)}$ bottom line from the law of total probability
Two-way table	joint = one cell ÷ grand total; marginal = a total ÷ grand total; conditional = one cell ÷ its row or column total

Your turn

Use the dice grid where it helps. Answers in grey.

Two dice

1. p(a double)
2. p(total is 5)
3. p(not a double)
4. p(red is even OR blue is 1)
5. p(total is 8 | red is 3)

1. 6/36 = 1/6 2. 4/36 = 1/9 3. 5/6 4. (18 + 6 − 3)/36 = 21/36 = 7/12 5. 1/6 (only blue 5)

Cards and tables

6. Two cards without replacement: p(both aces)?
7. Urn table: p(Small | Blue)?
8. Are 'coin shows heads' and 'die shows 6' independent? p(both)?

6. 4/52 × 3/51 = 1/221 7. 15/30 = 0.5 8. yes; 1/2 × 1/6 = 1/12

Bayes

9. A disease affects 2% of people. A test catches 90% of cases but also gives 5% false positives. Using 1000 people, find p(disease | positive).

9. 20 ill, 18 positive; 980 well, 49 positive; 18/67 ≈ 0.27

Find the unknown

10. p(A) = 0.6, p(B) = 0.5, p(A ∪ B) = 0.8. Find p(A ∩ B). Are A and B independent?
11. p(A | B) = 0.25 and p(B) = 0.4. Find p(A ∩ B).
12. p(A ∩ B) = 0.12 and p(A) = 0.3. Find p(B | A).
13. A and B are mutually exclusive, p(A) = 0.35, p(A ∪ B) = 0.8. Find p(B).
14. A and B are independent, p(A) = 0.4, p(A ∪ B) = 0.7. Find p(B).
15. 300 students: 40% come by bus; 25% of bus users are late, and 10% of the others. How many are late? p(bus | late)?

10. 0.6 + 0.5 − 0.8 = 0.3; 0.6 × 0.5 = 0.3, so independent 11. 0.4 × 0.25 = 0.1 12. 0.12 ÷ 0.3 = 0.4 13. 0.8 − 0.35 = 0.45 (no overlap) 14. 0.7 = 0.4 + p − 0.4p, so p = 0.5 15. bus 120, late 30; others 180, late 18; 48 late; 30/48 = 0.625